

Barcelona-Catalunya Grand Prix - Post-Race Strategy Report

Executive summary

This PDF is generated locally by the Python backend from the selected imported real-data bundle. It is designed as a portfolio-friendly export: readable, compact, and focused on the engineering story.

Driver battle

HAM had the stronger average clean pace by 0.727s per lap. HAM was more consistent, while RUS managed degradation better. The finishing advantage went to HAM (P1).

Selected-driver notes

HAM showed strong clean-air pace and was reasonably consistent once non-representative laps were removed. The best phase was stint 4 on H tyres, while stint 1 was the slowest phase. The degradation trend suggests the tyres reached their limit too quickly.

Rival notes

RUS showed steady clean-air pace and was reasonably consistent once non-representative laps were removed. The best phase was stint 3 on H tyres, while stint 1 was the slowest phase. Tyre management was acceptable, with room to smooth the long-run drop-off.

Full-grid race engineer ranking

Pos	Driver	Team	Pace(s)	Consistency	Tyres	Score
P2	RUS	Mercedes	82.369	69.0	67.1	75.8
P2	ANT	Mercedes	82.303	71.3	63.9	75.0
P1	HAM	Ferrari	81.642	71.3	43.5	73.7
P3	NOR	McLaren	82.381	69.0	60.5	73.3
P4	VER	Red Bull Racing	82.225	70.6	46.3	69.6
P6	LEC	Ferrari	82.544	70.1	54.9	69.4
P5	PIA	McLaren	82.735	71.0	57.9	69.1
P6	HAD	Red Bull Racing	83.264	63.4	64.2	62.8
P9	LAW	Racing Bulls	84.348	72.5	62.2	60.6
P12	HUL	Audi	84.651	82.3	46.0	58.1
P8	COL	Alpine	84.374	70.1	60.5	57.8
P10	LIN	Racing Bulls	84.455	58.4	76.3	55.5
P7	GAS	Alpine	84.182	64.8	53.4	53.8
P22	STR	Aston Martin	86.212	93.3	72.3	52.6
P12	SAI	Williams	84.662	67.5	50.1	50.6
P15	BEA	Haas F1 Team	84.942	61.8	61.4	49.9
P18	ALO	Aston Martin	86.621	74.3	65.4	46.7
P11	BOR	Audi	84.509	54.9	21.1	38.3
P14	PER	Cadillac	85.778	60.1	24.4	35.3
P11	ALB	Williams	85.005	58.3	0.0	29.7
P13	OCO	Haas F1 Team	85.054	59.2	0.0	29.4
P21	BOT	Cadillac	87.223	68.3	0.0	28.3

Data and modelling note

The project currently uses the imported real-data bundle when available. Pit laps, in-laps, out-laps, safety-car laps, and extreme outliers are removed from clean-lap calculations.

SciPy tyre degradation model

HAM on M: base 80.7387s, linear -0.01034, quadratic 0.006693.
RMSE: 0.1488s.

The fitted degradation trend is steep, suggesting a tyre drop-off; RMSE of 0.15s indicates a close fit.